

MPECI 2025/2026 — Practical Guide 1

Exam Study Notes + MATLAB Solutions Reference

1. BINOMIAL DISTRIBUTION (Bernoulli Series)

A **Binomial experiment** requires: (1) n independent trials, (2) each trial has only 2 outcomes (success/failure), (3) constant probability p per trial.

$$P(k) = C(n, k) \cdot p^k \cdot (1-p)^{n-k}$$

Symbol	Meaning	Example
n	Number of trials / tosses	$n = 15$ tosses
k	Number of successes desired	$k = 6$ heads
p	Probability of success per trial	$p = 0.5$ (fair coin)
$C(n, k)$	<code>nchoosek(n, k)</code> in MATLAB	$C(3, 2) = 3$

MATLAB — Binomial Calculation

```
prob = nchoosek(n, k) * p^k * (1-p)^(n-k); % exact formula
```

```
% Simulation: tosses = rand(n, N) < p; % n×N matrix, 1=success prob =  
sum(sum(tosses)==k) / N; % relative frequency
```

■ EXAM: 'At least k ' → use `sum(tosses)>=k` instead of `==k`

■ EXAM: Simulation result approaches analytical as $N \rightarrow \infty$ (Law of Large Numbers)

2. BER — Bit Error Rate (Task 1 Q5)

BER = probability of a single bit being in error. Bits are **independent**.

Packet of L bytes = $8L$ bits. Errors follow a Binomial distribution.

$$P(0 \text{ errors in } n \text{ bits}) = (1 - \text{BER})^n$$

$$P(\geq 2 \text{ errors}) = 1 - P(0 \text{ errors}) - P(1 \text{ error})$$

Case	Formula	Result
100B, $P(\text{no errors})$	$(1 - 10^{-6})^{800}$	≈ 0.99202
1000B, $P(\geq 2 \text{ errors})$	$1 - (1 - \text{BER})^{8000} - 8000 \cdot \text{BER} \cdot (1 - \text{BER})^{7999}$	≈ 0.00268

3. CONDITIONAL PROBABILITY & INDEPENDENCE

$$P(A|B) = P(A \cap B) / P(B)$$

$$A \perp B \text{ (independent)} \quad \blacksquare \quad P(A \cap B) = P(A) \cdot P(B) \quad \blacksquare \quad P(A|B) = P(A)$$

Classic Trap — Two Children Problem (T2 Q1)

Question	Conditioning Event	Sample Space	Answer
$P(2 \text{ boys} \mid \geq 1 \text{ boy})$	At least 1 is boy	{BB, BG, GB}	$1/3 \approx 0.333$
$P(2\text{nd boy} \mid 1\text{st is boy})$	First child is boy	{BB, BG}	$1/2 = 0.500$
$P(\geq 1 \text{ boy})$	None	{BB, BG, GB, GG}	$3/4 = 0.750$

■ **KEY EXAM POINT:** These two questions give different answers (1/3 vs 1/2) because the conditioning event is different! Do not confuse them.

MATLAB — Conditional Probability Simulation

```
% P(A|B) by simulation: prob_A_given_B = sum(A & B) / sum(B);
```

4. TOTAL PROBABILITY & BAYES THEOREM

Used when you know $P(B|\text{cause})$ but want $P(\text{cause}|B)$ — *inverting the direction of conditioning*.

$$P(B) = \sum P(B|A_i) \cdot P(A_i) \quad [\text{Total Probability}]$$

$$P(A_i|B) = P(B|A_i) \cdot P(A_i) / P(B) \quad [\text{Bayes Theorem}]$$

Wireless Link Example (T2 Q6) — worked solution

	P(condition)	P(error condition)
Normal link (N)	0.98	0.001
Interference (I)	0.02	0.10

$$P(\text{error}) = 0.001 \times 0.98 + 0.10 \times 0.02 = 0.00098 + 0.002 = 0.00298$$

$$P(\text{Normal}|\text{error}) = 0.00098 / 0.00298 = 0.3289$$

$$P(\text{Interf.}|\text{error}) = 0.00200 / 0.00298 = 0.6711$$

■ **INTERPRETATION:** Although interference is rare (2%), once an error occurs the link was under interference with 67% probability! This is a classic Bayes 'rare event amplification' effect.

5. BIRTHDAY PROBLEM / DART PROBLEM (T2 Q2, Q4)

Both problems share the same structure: assigning n items uniformly to m slots — what is $P(\text{at least one collision})$?

$$P(\text{no collision}) = (m/m) \cdot (m-1)/m \cdot (m-2)/m \cdot \dots \cdot (m-n+1)/m$$

$$P(\geq 1 \text{ collision}) = 1 - P(\text{no collision})$$

Problem	m (slots)	n	$P(\geq 1 \text{ collision})$
Birthday: >50%	365 days	23 people	> 0.50
Birthday: >90%	365 days	41 people	> 0.90
Darts n=20	100 targets	20 darts	≈ 0.87

```
% MATLAB: check for collisions hits = randi(m, 1, n); noCollision =
(numel(unique(hits)) == n); % Analytical: pNoCollision = prod((m-n+1:m) / m);
```

6. INDEPENDENCE — TWO DICE (T2 Q5)

Roll two fair dice. Sample space: 36 equally likely outcomes.

Event	Definition	Count	P(event)	Indep. with...
A	Sum = 9	4	4/36 = 1/9	B? YES
B	2nd die even	18	18/36 = 1/2	A? YES
C	At least one 5	11	11/36	D? NO
D	No 1s	25	25/36	C? NO

```
% Check A&B;: P(A∩B) = P(A)·P(B)? % A∩B = {sum=9 AND 2nd even} = {(3,6), (5,4)} → 2/36
% P(A)·P(B) = (4/36)·(18/36) = 72/1296 = 2/36 ✓ → INDEPENDENT
% C∩D = {at least one 5 AND no 1s} = 9/36 = 0.25 % P(C)·P(D) = (11/36)·(25/36) = 275/1296 ≈ 0.212 ≠ 0.25 → NOT INDEP.
```

7. KEY MATLAB COMMANDS — QUICK REFERENCE

Command	What it does
<code>rand(n,N)</code>	$n \times N$ matrix of uniform random values in $[0,1)$
<code>rand(n,N) < p</code>	Logical matrix: 1 with prob p (simulate Bernoulli)
<code>randi(m, 1, n)</code>	Row vector of n integers from 1 to m (uniform)
<code>sum(A)</code>	Column sums of matrix A (or sum of vector)
<code>sum(A, 2)</code>	Row sums of matrix A
<code>sum(v)==k</code>	Logical: 1 where v equals k
<code>sum(v)>=k</code>	Logical: 1 where v is at least k
<code>mean(v)</code>	Equivalent to $\text{sum}(v)/\text{length}(v)$ — use for probabilities!
<code>nchoosek(n,k)</code>	Binomial coefficient $C(n,k)$
<code>numel(unique(x))</code>	Count distinct values in x (collision check)
<code>histogram(v, edges, "Normalization", ProbabilityHistogram)</code>	Probability histogram
<code>stem(x,y)</code>	Discrete plot (for PMF)
<code>subplot(r,c,i)</code>	Create $r \times c$ grid of plots, activate i -th

8. EXAM TIPS & COMMON MISTAKES

- **Simulation vs Analytical:** Both should give the same answer. Difference decreases as N increases. Use $N \geq 10^4$ for 2-3% accuracy, $N = 10^5$ or 10^6 for high accuracy.
- **'At least k':** Always write `sum(tosses) >= k`. Analytically: $\sum P(k), P(k+1), \dots, P(n)$ OR $1 - P(0) - \dots - P(k-1)$.
- **Bytes vs Bits:** 1 byte = 8 bits. For BER problems, multiply packet size in bytes by 8.
- **Conditional probability direction:** $P(A|B) \neq P(B|A)$. Always divide by the probability of what you know.
- **Independence test:** $P(A \cap B) = P(A) \cdot P(B)$? If yes $\rightarrow$ independent. Equivalently: $P(A|B) = P(A)$?
- **Birthday problem:** Answer is $n=23$ for $>50\%$, $n=41$ for $>90\%$. Memorize these! They are classic exam questions.
- **Bayes 'rare event':** Even rare causes can become the dominant explanation when the effect (error) is observed. Always compute $P(B)$ first, then apply Bayes.
- **Function parameters:** When writing the `coinTossSim` function, include: p (success prob), $nTosses$, $kHeads$, N (simulations). This function is reusable for ALL Binomial problems.